

Fall 2025 CPTS530 Homework 4  
MATH 448-548  
Numerical Analysis

Aryan Ritwajeet Jha

October 2025

## Contents

|                                              |           |
|----------------------------------------------|-----------|
| <b>Problem 1</b>                             | <b>2</b>  |
| <b>Problem 2</b>                             | <b>4</b>  |
| <b>Problem 3</b>                             | <b>6</b>  |
| <b>Problem 4</b>                             | <b>7</b>  |
| <b>Problem 5</b>                             | <b>8</b>  |
| <b>Problem 6</b>                             | <b>9</b>  |
| Problem 6(a) . . . . .                       | 9         |
| Problem 6(b) . . . . .                       | 10        |
| Problem 6(c) . . . . .                       | 11        |
| <b>Problem 7</b>                             | <b>12</b> |
| <b>A Matrix Geometric Series</b>             | <b>13</b> |
| <b>B Determinant Properties</b>              | <b>14</b> |
| B.1 Determinant of Transpose . . . . .       | 14        |
| B.2 Determinant of Scalar Multiple . . . . . | 15        |
| <b>References</b>                            | <b>16</b> |

## Problem 1

### Problem Statement (Textbook 4.1.7):

Let  $A$  have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are  $n \times n$ . Prove that if  $(B - I)$  is nonsingular, then, for  $k \geq 1$ ,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$


---

### Solution:

To understand the pattern, let's first compute  $A^2$  and  $A^3$  directly.

#### Computing $A^2$ :

$$\begin{aligned} A^2 &= \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \end{aligned}$$

#### Computing $A^3$ :

$$\begin{aligned} A^3 &= A^2 \cdot A = \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^3 & B^2C + BC + C \\ 0 & I \end{bmatrix} \end{aligned}$$

Observing the pattern in the upper-right block:

$$\begin{aligned} k = 1 : & \quad C \\ k = 2 : & \quad BC + C \\ k = 3 : & \quad B^2C + BC + C \end{aligned}$$

We can factor these as:

$$\begin{aligned} k = 1 : & \quad (I)C \\ k = 2 : & \quad (B + I)C \\ k = 3 : & \quad (B^2 + B + I)C \end{aligned}$$

This suggests the general formula: the upper-right block is  $(I + B + B^2 + \cdots + B^{k-1})C$ , which equals  $(B^k - I)(B - I)^{-1}C$  (see Appendix A).

Now we prove this result by induction on  $k$ .

**Base Case ( $k = 1$ ):**

For  $k = 1$ , we need to verify that

$$A^1 = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} = \begin{bmatrix} B^1 & (B^1 - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

This simplifies to showing that  $C = (B - I)(B - I)^{-1}C$ , which is trivially true since  $(B - I)(B - I)^{-1} = I$ .

**Inductive Step:**

Assume the formula holds for  $k = n$ :

$$A^n = \begin{bmatrix} B^n & (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

We need to prove it holds for  $k = n + 1$ . Computing  $A^{n+1} = A^n \cdot A$ :

$$\begin{aligned} A^{n+1} &= \begin{bmatrix} B^n & (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^n \cdot B + (B^n - I)(B - I)^{-1}C \cdot 0 & B^n \cdot C + (B^n - I)(B - I)^{-1}C \cdot I \\ 0 \cdot B + I \cdot 0 & 0 \cdot C + I \cdot I \end{bmatrix} \\ &= \begin{bmatrix} B^{n+1} & B^n C + (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix} \end{aligned}$$

Now we need to show that the upper-right block equals  $(B^{n+1} - I)(B - I)^{-1}C$ .

**Key Identity:**

We claim that  $(B^{n+1} - I)(B - I)^{-1}C = B^n C + (B^n - I)(B - I)^{-1}C$ .

This identity is equivalent to showing that:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \dots + B^{k-1}$$

The proof of this geometric series identity for matrices is provided in Appendix A.

Thus, we have shown that:

$$A^{n+1} = \begin{bmatrix} B^{n+1} & (B^{n+1} - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

By the principle of mathematical induction, the formula holds for all  $k \geq 1$ .

---

## Problem 2

### Problem Statement (Textbook 4.1.12):

Let  $A$  be an  $n \times n$  invertible matrix, and let  $u$  and  $v$  be two vectors in  $\mathbb{R}^n$ . Find the necessary and sufficient conditions on  $u$  and  $v$  in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

**Solution:** (Approach followed as per lecture notes [1])

Let  $\tilde{A} = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$  where  $A$  is  $n \times n$  invertible, and  $u, v \in \mathbb{R}^n$ .

### Part 1: Necessary and Sufficient Condition for Invertibility

For  $\tilde{A}$  to be invertible, we need  $\tilde{A}x = 0 \Rightarrow x = 0$ .

Write  $x = \begin{bmatrix} \bar{x} \\ x_n \end{bmatrix}$  where  $\bar{x} \in \mathbb{R}^n$  and  $x_n \in \mathbb{R}$ . Then:

$$\tilde{A}x = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix} \begin{bmatrix} \bar{x} \\ x_n \end{bmatrix} = \begin{bmatrix} A\bar{x} + x_n u \\ v^T \bar{x} \end{bmatrix} = 0$$

This gives us:

$$A\bar{x} + x_n u = 0 \tag{1}$$

$$v^T \bar{x} = 0 \tag{2}$$

From (1):  $\bar{x} = -x_n A^{-1}u$  (since  $A$  is invertible).

Substituting into (2):  $v^T(-x_n A^{-1}u) = 0$ , which gives  $-x_n(v^T A^{-1}u) = 0$ .

If  $s := v^T A^{-1}u \neq 0$ , then  $x_n = 0$ , and from (1),  $A\bar{x} = 0 \Rightarrow \bar{x} = 0$  (since  $A$  is invertible).

Therefore,  $\tilde{A}$  is invertible if and only if:

$$\boxed{s = v^T A^{-1}u \neq 0}$$

### Part 2: Formula for the Inverse

Seek  $\tilde{A}^{-1} = \begin{bmatrix} B & \hat{u} \\ \hat{v}^T & k \end{bmatrix}$  where  $B \in \mathbb{R}^{n \times n}$ ,  $\hat{u}, \hat{v} \in \mathbb{R}^n$ ,  $k \in \mathbb{R}$ .

From  $\tilde{A}\tilde{A}^{-1} = I_{n+1}$ , block multiplication gives:

$$AB + u\hat{v}^T = I_n \tag{3}$$

$$A\hat{u} + ku = 0 \tag{4}$$

$$v^T B = 0^T \tag{5}$$

$$v^T \hat{u} = 1 \tag{6}$$

From (4):  $\hat{u} = -kA^{-1}u$ . Substituting into (6):

$$v^T(-kA^{-1}u) = 1 \Rightarrow -ks = 1 \Rightarrow k = -\frac{1}{s}$$

Therefore:  $\hat{u} = -kA^{-1}u = \frac{1}{s}A^{-1}u$ .

From (3):  $B = A^{-1}(I_n - u\hat{v}^T)$ . Substituting into (5):

$$v^T A^{-1}(I_n - u\hat{v}^T) = 0^T$$

$$v^T A^{-1} - (v^T A^{-1}u)\hat{v}^T = 0^T$$

$$v^T A^{-1} - s\hat{v}^T = 0^T$$

$$\hat{v}^T = \frac{1}{s}v^T A^{-1}$$

Therefore:  $\hat{v} = \frac{1}{s}(A^{-1})^T v = \frac{1}{s}(A^T)^{-1}v$ .

Finally, from  $B = A^{-1}(I_n - u\hat{v}^T)$ :

$$B = A^{-1} \left( I_n - u \cdot \frac{1}{s}v^T A^{-1} \right) = A^{-1} - \frac{1}{s}A^{-1}uv^T A^{-1}$$

**Final Answer:**

When  $s = v^T A^{-1}u \neq 0$ :

$$\tilde{A}^{-1} = \begin{bmatrix} A^{-1} - \frac{1}{s}A^{-1}uv^T A^{-1} & \frac{1}{s}A^{-1}u \\ \frac{1}{s}v^T A^{-1} & -\frac{1}{s} \end{bmatrix}$$

where  $s = v^T A^{-1}u$ .

---

## Problem 3

### Problem Statement (Textbook 4.1.17):

A square matrix  $A$  is said to be **skew-symmetric** if  $A^T = -A$ . Prove that if  $A$  is skew-symmetric, then  $x^T Ax = 0$  for all  $x$ .

---

### Solution:

Let  $A$  be a skew-symmetric matrix, so  $A^T = -A$ . Consider the scalar quantity  $x^T Ax$ .

Taking the transpose of this scalar (which equals itself):

$$\begin{aligned}(x^T Ax)^T &= x^T A^T (x^T)^T \\ &= x^T A^T x \\ &= x^T (-A)x \\ &= -x^T Ax\end{aligned}$$

Since  $x^T Ax$  is a scalar and equals its own negative, we have  $x^T Ax = -x^T Ax$ , which implies  $2x^T Ax = 0$ .

Therefore,  $x^T Ax = 0$  for all  $x$ . □

---

## Problem 4

### Problem Statement (Textbook 4.1.18):

(Continuation) Prove that the diagonal elements of a skew-symmetric matrix are **0**. Also, prove that the determinant is 0 when the matrix is of odd order.

#### Solution:

##### Part 1: Diagonal elements are zero.

Consider a general skew-symmetric matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \quad A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}$$

Since  $A^T = -A$ , comparing diagonal elements:

$$a_{11} = -a_{11}, \quad a_{22} = -a_{22}, \quad \dots, \quad a_{nn} = -a_{nn}$$

For any diagonal element  $a_{ii}$ , we have  $a_{ii} = -a_{ii}$ , which implies:

$$a_{ii} = (A^T)_{ii} = (-A)_{ii} = -a_{ii}$$

This implies  $2a_{ii} = 0$ , so  $a_{ii} = 0$  for all  $i$ .

##### Part 2: Determinant is zero for odd order.

**Examples:** Consider

$$A_3 = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{bmatrix} \text{ (odd order 3), } \quad A_2 = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix} \text{ (even order 2)}$$

We have  $\det(A_3) = 0$  while  $\det(A_2) = 25 \neq 0$ . This illustrates that the determinant is zero for odd order but not necessarily for even order.

#### General Proof:

For a skew-symmetric matrix  $A$  of order  $n$ , we have  $A^T = -A$ . Taking determinants:

$$\det(A^T) = \det(-A) = (-1)^n \det(A)$$

where we used  $\det(\lambda A) = \lambda^n \det(A)$  with  $\lambda = -1$  (see Appendix B.2).

Since  $\det(A^T) = \det(A)$  (see Appendix B.1), we get:

$$\det(A) = (-1)^n \det(A)$$

If  $n$  is odd, then  $(-1)^n = -1$ , giving  $\det(A) = -\det(A)$ , which implies  $\det(A) = 0$ .  $\square$

## Problem 5

### Problem Statement (Textbook 4.1.19):

(Continuation) Let  $A$  be any square matrix, and define  $A_0 = \frac{1}{2}(A + A^T)$  and  $A_1 = \frac{1}{2}(A - A^T)$ . Prove that  $A_0$  is symmetric, that  $A_1$  is skew-symmetric, that  $A = A_0 + A_1$  and that for all  $x$ ,  $x^T A x = x^T A_0 x$ . This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

---

**Solution:**

---



## Problem 6

### Part (a)

**Problem Statement (Textbook 4.2.1a):**

Prove these facts, needed in the proof of Theorem 2.

**Problem 6(a)** If  $U$  is upper triangular and invertible, then  $U^{-1}$  is upper triangular.

---

**Solution:**

---

**Part (b)**

**Problem Statement (Textbook 4.2.1b):**

Prove these facts, needed in the proof of Theorem 2.

**Problem 6(b)** The inverse of a unit lower triangular matrix is unit lower triangular.

---

**Solution:**

---

**Part (c)**

**Problem Statement (Textbook 4.2.1c):**

Prove these facts, needed in the proof of Theorem 2.

**Problem 6(c)** The product of two upper (lower) triangular matrices is upper (lower) triangular.

---

**Solution:**

---

## Problem 7

**Problem Statement (Textbook 4.2.5):**

Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

---

**Solution:**

---

## A Matrix Geometric Series

### Lemma

For any square matrix  $B$  such that  $(B - I)$  is nonsingular, and for any  $k \geq 1$ :

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \cdots + B^{k-1}$$

### Proof:

Define the partial sum sequence:

$$\begin{aligned} S_0 &= 0 \\ S_1 &= I \\ S_2 &= I + B \\ S_3 &= I + B + B^2 \\ &\vdots \\ S_k &= I + B + B^2 + \cdots + B^{k-1} \end{aligned}$$

We will show that  $(B - I)S_k = B^k - I$  for all  $k \geq 1$ .

Multiplying  $S_k$  by  $(B - I)$ :

$$\begin{aligned} (B - I)S_k &= (B - I)(I + B + B^2 + \cdots + B^{k-1}) \\ &= B(I + B + B^2 + \cdots + B^{k-1}) - (I + B + B^2 + \cdots + B^{k-1}) \\ &= (B + B^2 + B^3 + \cdots + B^k) - (I + B + B^2 + \cdots + B^{k-1}) \\ &= B^k - I \end{aligned}$$

Since  $(B - I)$  is nonsingular by assumption, we can multiply both sides by  $(B - I)^{-1}$  on the left:

$$S_k = (B^k - I)(B - I)^{-1}$$

Therefore:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \cdots + B^{k-1}$$

This completes the proof. □

## B Determinant Properties

### B.1 Determinant of Transpose

#### Theorem

For any square matrix  $A$ ,  $\det(A^T) = \det(A)$ .

#### Examples:

For a  $2 \times 2$  matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Direct computation:

$$\det(A) = ad - bc, \quad \det(A^T) = ad - cb = ad - bc$$

For a  $3 \times 3$  matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

Expanding  $\det(A)$  along the first row:

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Expanding  $\det(A^T)$  along the first column:

$$\det(A^T) = a_{11} \begin{vmatrix} a_{22} & a_{32} \\ a_{23} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{31} \\ a_{23} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{31} \\ a_{22} & a_{32} \end{vmatrix}$$

Each  $2 \times 2$  minor in  $\det(A^T)$  is the transpose of the corresponding minor in  $\det(A)$ , and by the  $2 \times 2$  case, they have equal determinants. Thus  $\det(A^T) = \det(A)$ .

#### General Proof (by Induction):

**Base case:** For  $n = 1$ ,  $A = [a]$  and  $A^T = [a]$ , so  $\det(A) = \det(A^T) = a$ . For  $n = 2$ , verified above.

**Inductive step:** Assume  $\det(B^T) = \det(B)$  for all  $(n-1) \times (n-1)$  matrices  $B$ .

For an  $n \times n$  matrix  $A$ , expand  $\det(A)$  along row  $i$ :

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij} = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$$

where  $M_{ij}$  is the  $(n-1) \times (n-1)$  minor obtained by deleting row  $i$  and column  $j$ . For  $A^T$ , the element in position  $(j, i)$  is  $a_{ij}$ . Expanding  $\det(A^T)$  along column  $i$ :

$$\det(A^T) = \sum_{j=1}^n a_{ij} C_{ji}^T = \sum_{j=1}^n (-1)^{j+i} a_{ij} M_{ji}^T$$

The minor  $M_{ji}^T$  (obtained by deleting row  $j$  and column  $i$  from  $A^T$ ) is the transpose of  $M_{ij}$ . By the inductive hypothesis,  $\det(M_{ji}^T) = \det(M_{ij})$ .

Since  $(-1)^{i+j} = (-1)^{j+i}$ , we have:

$$\det(A^T) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(M_{ij}) = \det(A)$$

Therefore, by induction,  $\det(A^T) = \det(A)$  for all  $n \times n$  matrices.  $\square$

## B.2 Determinant of Scalar Multiple

### Theorem

For any  $n \times n$  matrix  $A$  and scalar  $\lambda$ ,  $\det(\lambda A) = \lambda^n \det(A)$ .

### Proof:

When we multiply a matrix  $A$  by a scalar  $\lambda$ , each entry becomes  $\lambda a_{ij}$ . Using the multilinearity property of determinants, we can factor out  $\lambda$  from each of the  $n$  rows (or columns):

$$\det(\lambda A) = \det \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{n1} & \lambda a_{n2} & \cdots & \lambda a_{nn} \end{bmatrix}$$

Factoring  $\lambda$  from the first row:

$$= \lambda \det \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{n1} & \lambda a_{n2} & \cdots & \lambda a_{nn} \end{bmatrix}$$

Repeating this process for all  $n$  rows:

$$= \lambda^n \det \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \lambda^n \det(A)$$

Therefore,  $\det(\lambda A) = \lambda^n \det(A)$ .  $\square$

## References

- [1] Alexander Panchenko. Math 448 lecture notes. Washington State University, Fall 2025. Pages 93-130.